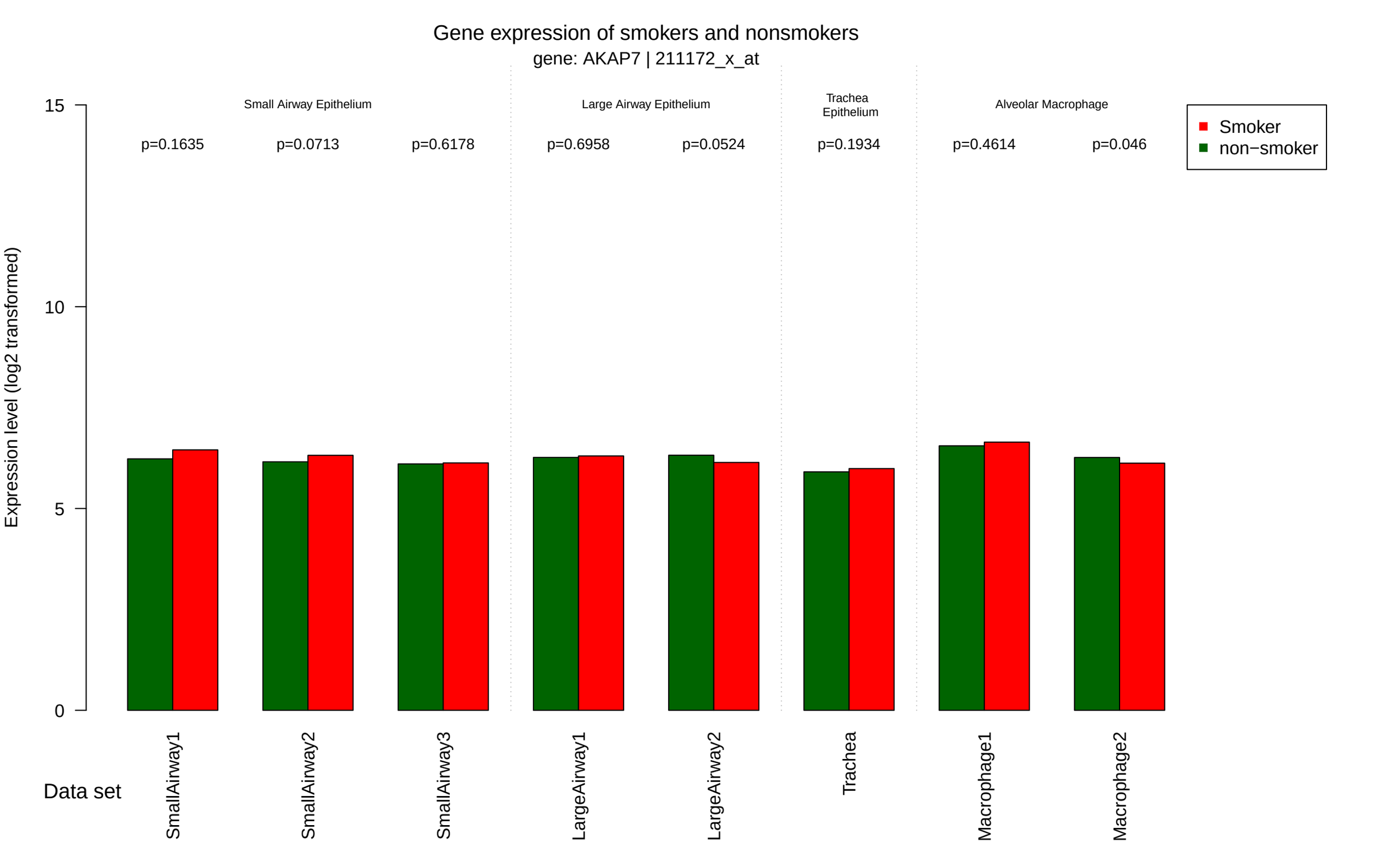
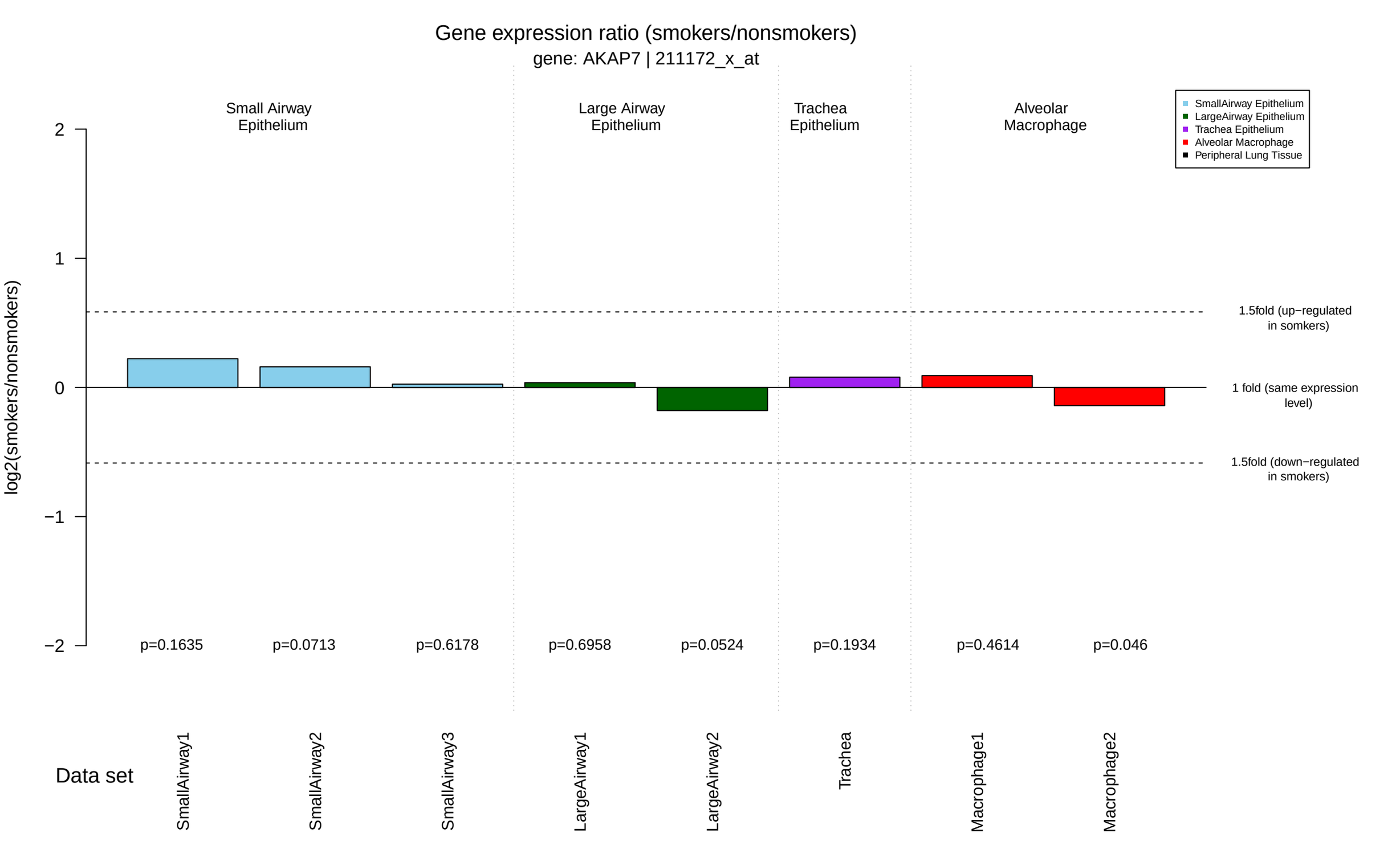
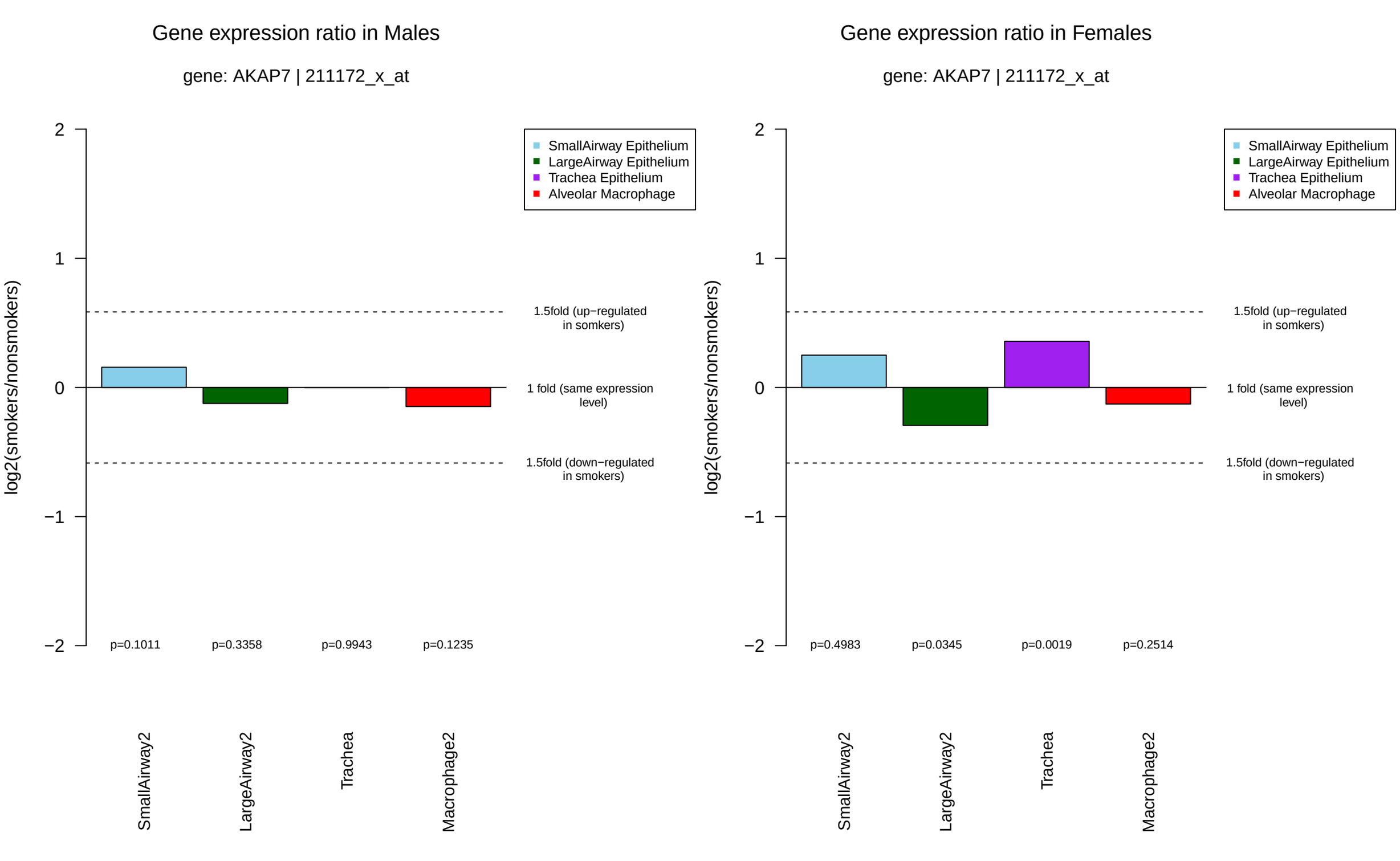
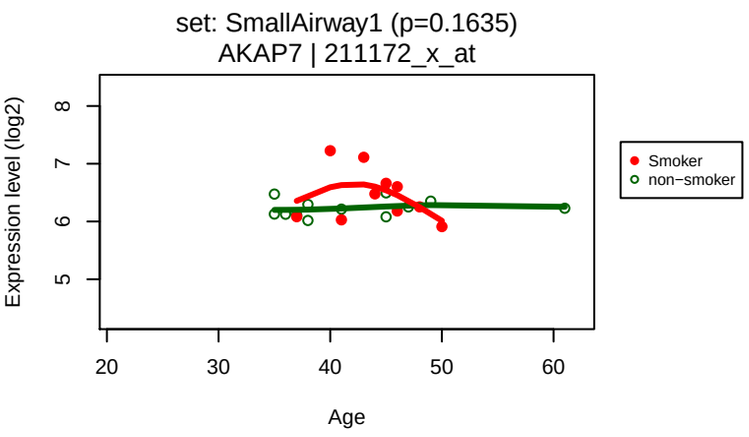
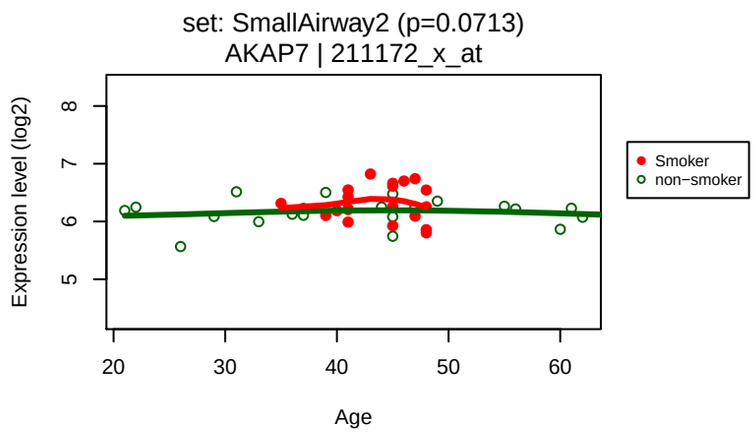
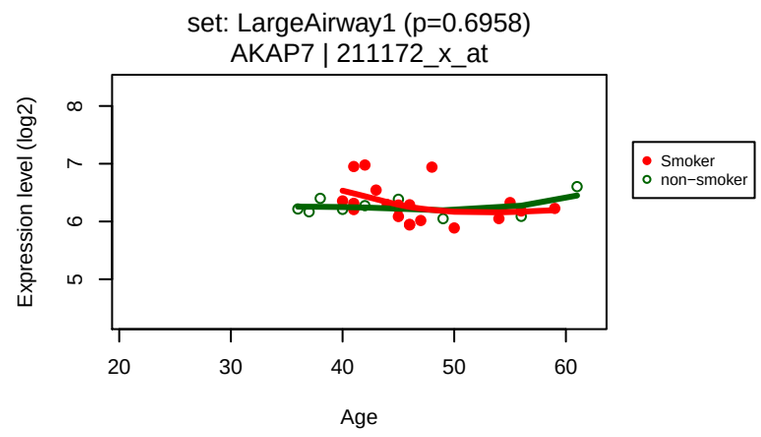

Gene expression of smokers and nonsmokers
gene: AKAP7 | 211172_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1635
p=0.0713
p=0.6178
p=0.6958
p=0.0524
p=0.1934
p=0.4614
p=0.046
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

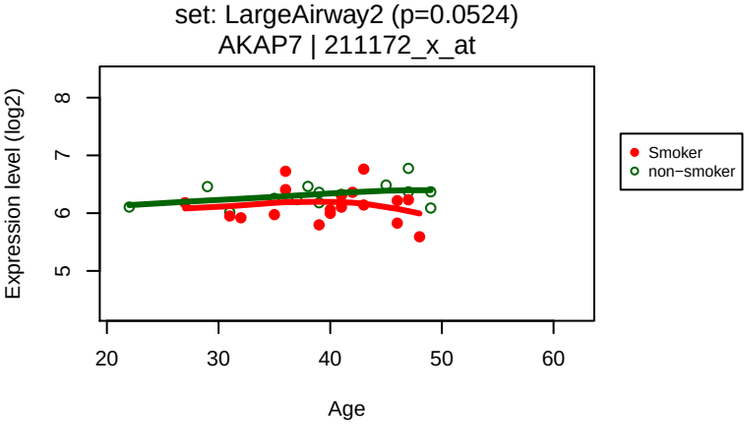

Gene expression ratio (smokers/nonsmokers)
gene: AKAP7 | 211172_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1635
p=0.0713
p=0.6178
p=0.6958
p=0.0524
p=0.1934
p=0.4614
p=0.046
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

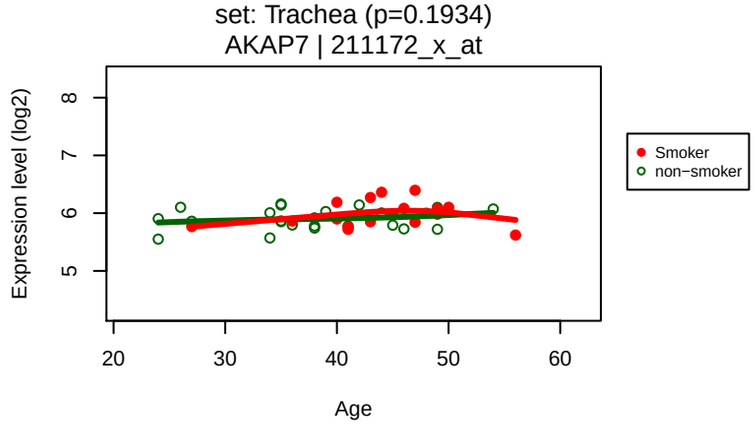

Gene expression ratio in Males
gene: AKAP7 | 211172_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1011
p=0.3358
p=0.9943
p=0.1235
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP7 | 211172_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4983
p=0.0345
p=0.0019
p=0.2514
SmallAirway2
LargeAirway2
Trachea
Macrophage2

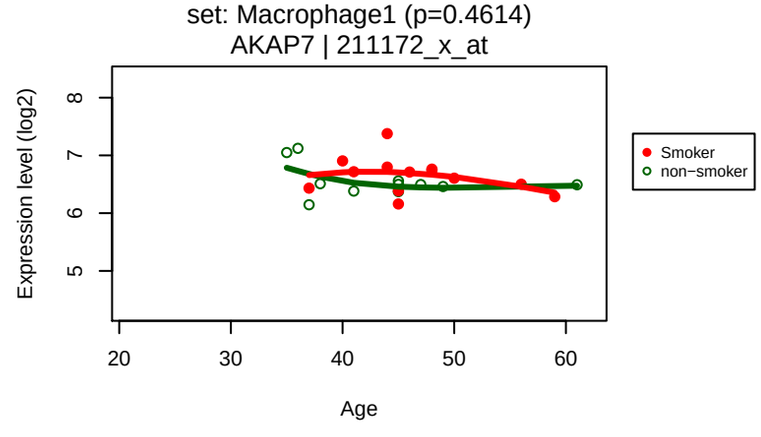

set: SmallAirway1 (p=0.1635)
AKAP7 | 211172_x_at
Expression level (log2)
Age
Smoker
non−smoker

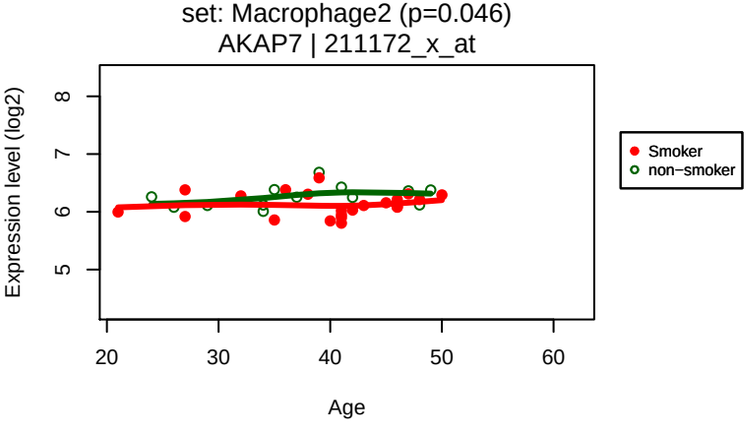

set: SmallAirway2 (p=0.0713)
AKAP7 | 211172_x_at
Expression level (log2)
Age
Smoker
non−smoker

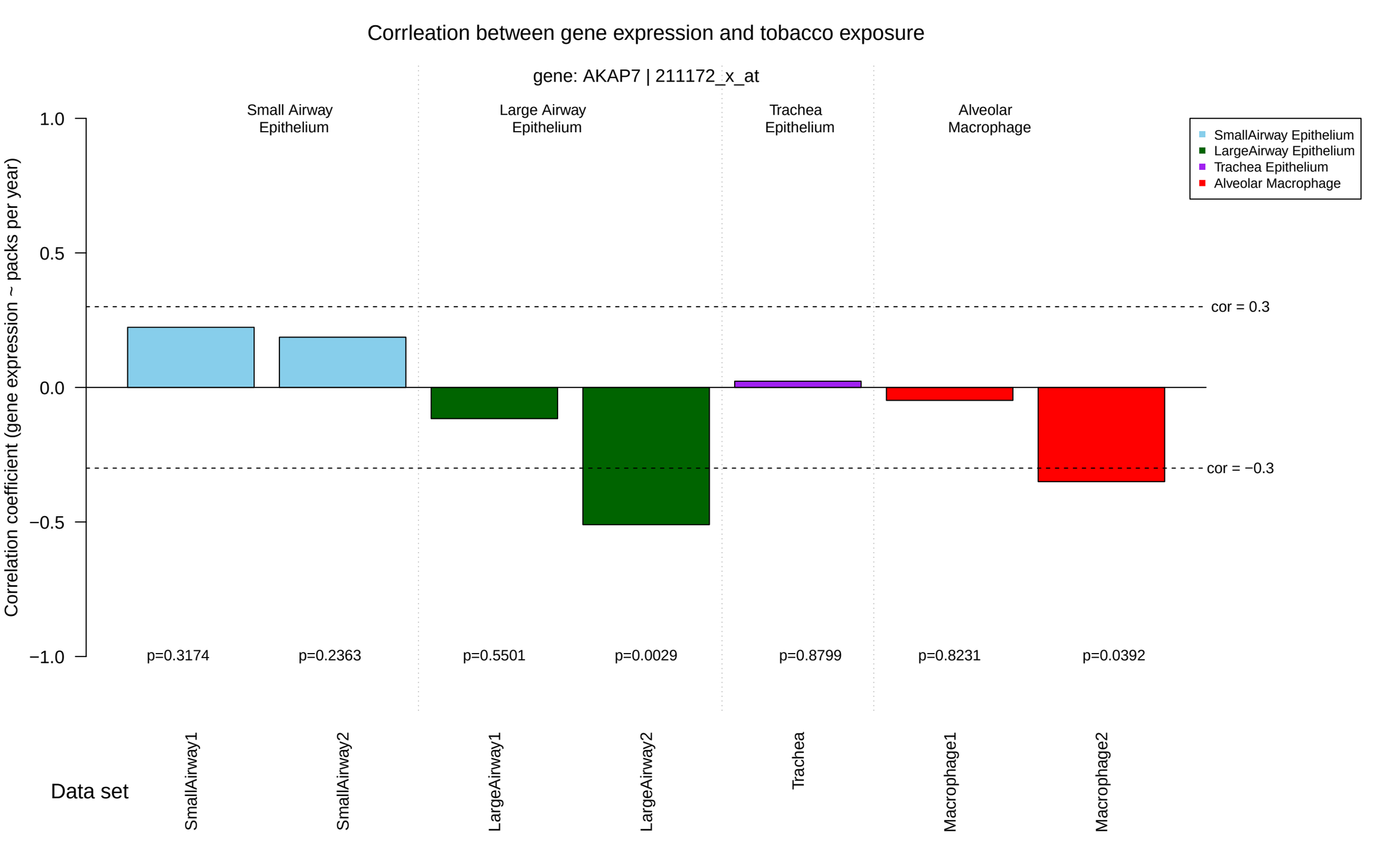

set: LargeAirway1 (p=0.6958)
AKAP7 | 211172_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0524)
AKAP7 | 211172_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1934)
AKAP7 | 211172_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4614)
AKAP7 | 211172_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.046)
AKAP7 | 211172_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP7 | 211172_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3174
p=0.2363
p=0.5501
p=0.0029
p=0.8799
p=0.8231
p=0.0392
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2